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MNE-RT: A Python package for real-time M/EEG neurofeedback and brain-computer interfaces

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DOI: 10.xxxxxx/draft

Software

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Editor:

Submitted: 13 October 2026
Published: unpublished

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Summary

In neurofeedback and brain-computer interfacing (BCI), acquisition, artifact handling, feature extraction, decision and display must all complete within a single analysis window, typically a few hundred milliseconds.

`MNE-RT` is a Python package that implements this processing chain on top of `MNE-Python` (Gramfort et al., 2013) and `MNE-LSL` (Scheltienne et al., 2025). A session is expressed as a single object. `RTStream` connects to a Lab Streaming Layer (LSL) source (Kothe et al., 2025), records a resting-state baseline, and then runs a loop in which each window is cleaned, reduced to one or more neural features, standardised against that baseline, evaluated by an adaptive reward protocol, and published to live displays, an LSL outlet and OSC receivers. Sessions are written to disk in a BIDS-compatible layout (Pernet et al., 2019) together with per-window latency, artifact-rate and signal-to-noise records, so that a session can be audited or replayed offline through the same code path.

The package provides 20 feature modalities spanning sensor and source space, 10 adaptive protocols, 5 online artifact-correction methods, 9 live visualisation windows, and an `mne-rt` command-line interface. Three hardware-free acquisition paths (replay of any MNE-readable recording through a mock LSL stream, streaming from an in-memory array, and the bundled `mne-rt demo`) allow the pipeline to be run, taught and tested without an amplifier.

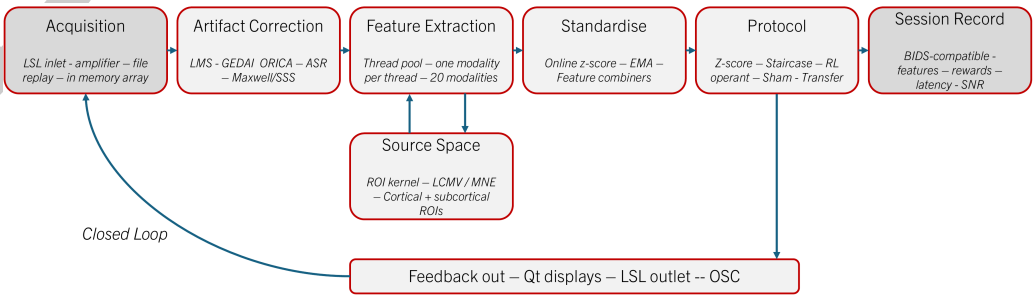


Figure 1: The `MNE-RT` closed loop. Data arrive over LSL, are corrected in place, reduced to features by a thread pool, standardised and combined, evaluated by an adaptive protocol, and returned to the participant through displays, an LSL outlet or OSC.

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Statement of need

Real-time M/EEG stacks are commonly assembled per study (Sitaram et al., 2017), combining streaming, artifact rejection, feature extraction, adaptive thresholding, stimulus communication and logging from separate components. The resulting configuration is frequently not recorded with the data: the thresholding rule, the achieved latency and the artifact method may exist only in a local script.

`MNE-RT` is intended for researchers who already analyse M/EEG with `MNE-Python` and who want the online experiment expressed with the same objects, with the protocol, the artifact method and the measured timing stored alongside the recorded session.

Sensor-space band power is already well supported by existing real-time tools, so `MNE-RT` does not add a new method there. Source-space features are the gap: the implementations available today are not fast enough to run inside a real-time loop, and this is what `MNE-RT` contributes a method for. Training a region-to-region interaction, for example imaginary coherency (Nolte et al., 2004) between a pair of regions of interest reconstructed with an LCMV beamformer (Van Veen et al., 1997), requires projecting each window into source space and reducing thousands of grid points to a small number of ROI time courses. Measured on a 5 mm whole-brain volume source space (14,629 points), the direct `MNE-Python` implementation costs 650 to 800 ms per window against a 500 ms hop, so the loop does not close in real time. A volume source space also permits subcortical labels to be specified, although the spatial resolution attainable from a scalp montage limits the extent to which deep sources can be separated from overlying cortex.

State of the field

Within the MNE ecosystem, `mne-realtime` was the original real-time extension and is no longer maintained. `MNE-LSL` (Scheltienne et al., 2025) is actively maintained and provides modern LSL bindings, stream inlets and outlets, file replay and online epoching. `MNE-RT` depends on it as its acquisition backbone rather than duplicating it. `MNE-LSL` is scoped as a streaming library, whereas neurofeedback protocols, source-space feature kernels, reward logic and session provenance are application-level concerns outside that scope.

Outside the MNE ecosystem, `Timeflux` (Clisson et al., 2019) and `NeuXus` (Caetano et al., 2023) model real-time processing as a configurable graph of nodes; `MEDUSA` (Santamaría-Vázquez et al., 2023) provides a broad BCI ecosystem with its own paradigms and signal-processing stack; `OpenViBE` (Renard et al., 2010) and `BCI2000` (Schalk et al., 2004) are mature C++ platforms driven through graphical designers; NeuroPype is proprietary. Each of these requires users to adapt their data and processing pipeline to platform-specific data structures and toolchains, and none provides an anatomically constrained source-space feature path. `MNE-RT` occupies the remaining position: an MNE-native framework in which the online feature is expressed with the same objects, montages, forward models and inverse operators as the offline analysis that motivated it, and in which source-space features are computed within the real-time budget.

Software design

`MNE-RT` is script-shaped rather than graph-shaped. A session reads as `connect_to_lsl` → `record_baseline` → `record_main` → `save`, following the structure of an offline `MNE-Python` analysis rather than requiring the experiment to be expressed as a dataflow configuration. This design centralises multiple responsibilities within the session class, resulting in a larger and more complex class than a modular graph-based architecture would require. In return, an experiment can be expressed as a readable, version-controlled Python script that closely follows the structure of the corresponding offline analysis.

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Source-space features meet the real-time requirement by exploiting the linearity of the operators involved, rather than by accelerating the generic pipeline. LCMV with `pick_ori="max-power"` and minimum-norm variants with `pick_ori="normal"` are linear operators, and `mean`/`mean_flip` label extraction is a fixed sparse average. `SourceModel.roi_kernel()` therefore collapses whitening, inverse operator and label extraction into a single `(n_roi, n_channels)` matrix, applied per window as one matrix product. This reduces the figure quoted above to approximately 1 ms per window, and agrees with the `MNE-Python` result to better than 1e-10 relative error, which the test suite asserts directly. Free-orientation configurations are not linear, since taking the norm across three orientations discards phase, and the kernel is refused in that case rather than returning an incorrect value. The same precompute strategy applies to the rest of the loop: bad-channel detection, ICA, noise covariance, forward solution and the Maxwell/SSS basis (Taulu & Simola, 2006) are all constructed during the baseline phase, leaving a bounded per-window cost online. The head model is built lazily, so sensor-space sessions do not download anatomical data.

Everything that varies between studies is a strategy object behind a small interface. Protocols expose `evaluate(value) -> (crossed, magnitude)`, which keeps z-score, percentile, staircase (Levitt, 1971), reinforcement-learning, operant-schedule, multi-band, cross-session-transfer and double-blind sham protocols interchangeable; artifact correctors expose `fit`/`transform`, covering adaptive LMS, ORICA, GEDAI, ASR (Mullen et al., 2015) and real-time Maxwell filtering; feature combiners include one that accepts any scikit-learn estimator. Features themselves span band power and its derivatives (ERD/ERS (Pfurtscheller & Lopes da Silva, 1999), laterality, individual peak frequency via spectral parameterisation (Donoghue et al., 2020)), connectivity, cross-frequency coupling (Tort et al., 2010), graph learning (Kalofolias, 2016) and online decoding. A naming scheme (`base@label`) allows one measure to run concurrently in several bands with independent state and independent output channels.

Concurrency is explicit rather than framework-mediated. A daemon thread runs a 50%-overlap sliding window and fans modalities out across a thread pool, while the Qt main thread refreshes displays at 30 fps, so that visualisation cannot block acquisition. Window onsets are taken from the LSL clock and recorded as `NaN` rather than substituted when unavailable, which keeps traces alignable to a stimulus log. Interoperability is through standard protocols, an LSL outlet and OSC, rather than through a bundled stimulus engine, so `MNE-RT` can drive PsychoPy, Max/MSP or Unity without owning them.

Research impact statement

`MNE-RT` has been developed in the open since May 2024, comprising more than 500 commits and over 30 merged pull requests, and is released on both PyPI and conda-forge with versioned documentation, a tutorial series and an executable example gallery. Over 950 automated tests across 38 modules run in continuous integration on Python 3.11 to 3.13, including a job that resolves every dependency to its declared lower bound.

Its feature-extraction layer was presented at IEEE CBMS 2025 (Shabestari et al., 2025). Development has been supported by Swiss National Science Foundation grant 208164, *Advancing Neurofeedback in Tinnitus*, and the package is the online implementation vehicle for methods developed in that programme, including the holistic graph-learning network analysis reported by Shabestari et al. (2026). The source-space capability described above was developed for an ongoing clinical collaboration on neurofeedback for aphasia, in which region-to-region connectivity between language areas is trained in real time from a 64-channel EEG stream; that study is the first external deployment, and motivated the volume-source-space and connectivity work released in version 1.2.

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AI usage disclosure

Generative AI tools (Anthropic's Claude, used through Claude Code) assisted with parts of the implementation, the test suite, the documentation and the drafting of this manuscript. All AI-assisted output was reviewed, corrected and validated by the authors, who take full responsibility for the correctness, originality and licensing of the software and of this paper. Numerical claims in this paper were verified against the package's own test suite and measurements.

Acknowledgements

This work was supported by the Swiss National Science Foundation (grant 208164, *Advancing Neurofeedback in Tinnitus*). We thank the `MNE-Python` and `MNE-LSL` developer communities, on whose work this package depends.

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